

to the drugs being used in efforts to control the outbreak. The extent of the problem may be gauged from the fact that in the comparatively limited outbreak described, strains with eight different patterns of drug resistance, from full sensitivity to resistance to six different antimicrobial agents, were isolated, and that the interaction of at least three different R factors, conferring resistance to AST, SSu, and ASCTSu, respectively, is needed to explain the patterns of drug resistance observed, even if we accept that the R factor ASCSu may be a variant of the R factor ASCTSu. It cannot at present be decided whether the nalidixic-acid resistance is determined by a chromosomal mutation or by a plasmid.

The most economical hypothesis is that the outbreak started with the introduction of a fully sensitive strain of *Sh. flexneri* into the ward; that different lines of this strain acquired by conjugation different R factors already present in the faecal flora of the patients, in part under the selective pressure of the drugs used in attempts to control the outbreak; and that the use of nalidixic acid allowed the emergence of lines resistant to it. If drug treatment of an enteric infection is indicated, choice of the most effective drug should be based on the sensitivity of the normal faecal flora of the patient as well as on that of the pathogen.

Anderson and his colleagues have reported analyses of two strains of *S. typhimurium* of R type ASTSu showing striking differences in genetic behaviour. In the first, of phage type 1a (Anderson and Datta 1965), the resistance pattern ASTSu was transferred to K12 as a single unit and spontaneous loss of any part of the resistance spectrum implied loss of the whole spectrum. The behaviour was that of a single complex R factor. In the second strain, of phage-type 29, (Anderson and Lewis 1965a and b) the R type ASTSu was the result of the presence in the one strain of three basically independent resistance determinants, for A, SSu, and T, interacting with one transfer factor, the Δ factor, which could itself operate independently of these determinants. Walton and Fulton (1967) suggest that many workers are failing to appreciate the importance of the analysis of the segregation of different patterns of resistance among resistant progeny in simple mating experiments. Such analysis had in fact already been used by Anderson and Lewis (1965a and b) to demonstrate the different systems underlying the superficially similar R types of 1aASTSu and 29ASTSu. I describe here yet another variation on this theme, involving the presence of two complete R factors AST and SSu in one strain, each with its own transfer factor.

Finally, it must be stressed that in the laboratory investigation of epidemics of enteric disease it is inadequate to determine the antibiotic sensitivity patterns of the first few strains isolated and to assume that the later ones will be similar, and that antibiotic sensitivity patterns should be used with caution as epidemiological markers.

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REPLACEMENT OF AORTIC AND MITRAL VALVES WITH A PULMONARY AUTOGRAFT

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Summary The finding of late degenerative changes in a proportion of aortic homografts prompted the use of the patient's own pulmonary valve as a replacement. As a living autograft, the transplanted pulmonary valve has the prospect of long-term or permanent survival, whilst retaining the advantages of an aortic homograft. Similarly it can be used to replace the mitral valve. The excised pulmonary valve is replaced with a homograft.

Introduction

THE replacement of diseased aortic valves with homograft aortic valves is now well established (Ross 1964, Barratt Boyes et al. 1965, Hoeksema et al. 1967). Hubka et al. (1967) and Ionescu et al. (1967) have provided additional evidence that the inverted aortic homograft or heterograft valve will function as a substitute for the mitral valve, and I have used this method in 7 patients.

The advantages of the homograft valve are its freedom from embolic complications and the central flow orifice without obstructive turbulence and gradient which it offers. The main disadvantages, which are now becoming apparent, are a tendency to late failure (Smith 1967)—which may be related to the method of preparing the valve—and late calcification. Calcification is common in the aortic wall of the homograft, but it has also been seen as early as two years after operation in the cusps of some of the valves prepared with ethylene oxide and freeze-drying (Longmore et al. 1966).

In view of these findings and on the assumption that living autogenous tissue is the best available substitute in the body, the patient's own pulmonary valve has been excised and used to replace the aortic valve in 12 patients. The deficiency in the outflow from the right ventricle has been replaced with a homograft aortic valve, since this has been found to work satisfactorily in pulmonary atresia (Ross and Somerville 1966). In this area the pressure that closes the cusps is low, thus reducing the risk of trauma; hence late failure is unlikely to have the serious consequences of aortic-valve dysfunction. The hope is that the transplanted autograft in the aortic area will continue to function and survive indefinitely as an autograft nourished directly from the blood-stream. Similar considerations apply to replacement of the mitral valve.

Methods

Aortic Replacement

The aortic valve is excised in the ordinary way. Provided that there is no gross disproportion between the sizes of the aortic and pulmonary valve rings, a suitable aortic homograft is immersed in saline/penicillin solution and allowed to reconstitute during the subsequent excision of the pulmonary valve.

The pulmonary artery is transected just above the attachment of the pulmonary cusps and dissected from above down, keeping close to the wall of the pulmonary artery and away from the left main coronary artery which runs behind it. An additional short incision is made in the tract of the right-ventricular outflow just below the pulmonary-valve sinuses, and the inner muscle layer of the right ventricle is incised transversely around the base of the valve (fig. 1). At this point



Fig. 1—After transection of pulmonary-artery trunk, right ventricle and half the thickness of muscle of distal wall are incised transversely.

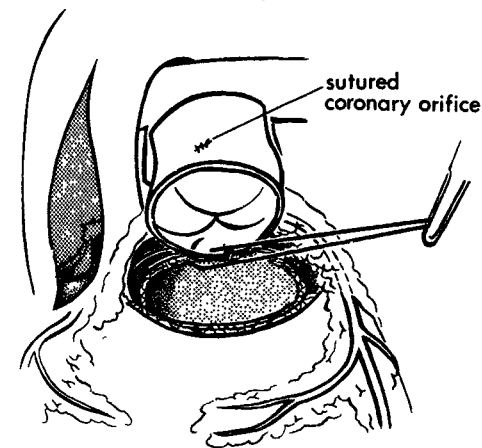


Fig. 2—Reconstituted aortic homograft is sewn proximally to cut edge of right ventricular muscle and distally to pulmonary-artery trunk.

the muscle is in two fairly clearly defined layers, and only the inner layer, together with the valve, are excised. At only one point posteriorly, where the pulmonary artery and aorta adhere closely, is there need for sharp division of the two vessels. In other areas, the plane of separation is easily defined. The plane of dissection and the operative field are made clearer by temporary occlusion of the coronary arteries. The excised pulmonary-valve bed is then inspected for bleeding venous sinuses. These are oversewn or lightly touched with a diathermy needle, care being taken not to injure the left main trunk of the coronary artery.

The excised pulmonary valve is left lying in blood in the aortic root or left ventricle during this period, after which the soft living muscle is trimmed from the bases of the sinuses.

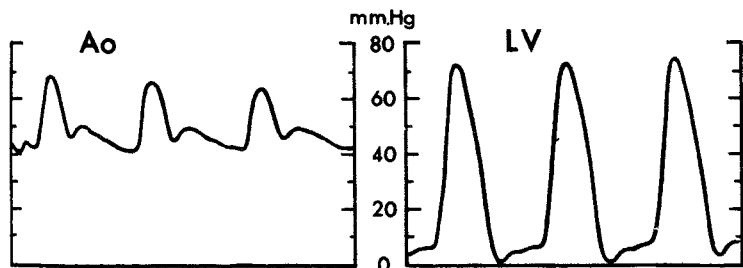


Fig. 5—Postoperative pressures after pulmonary autograft replacement of aortic valve in calcific aortic stenosis.

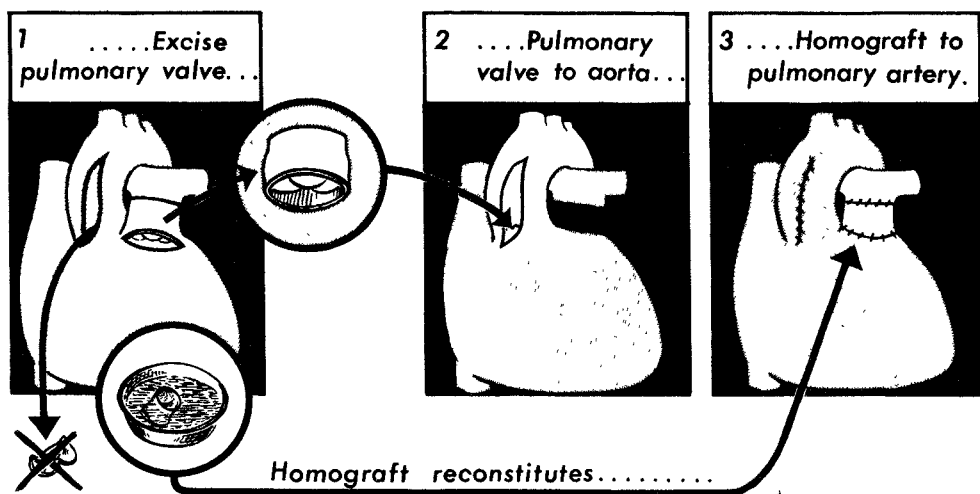


Fig. 3—Steps in replacement of aortic valve with a pulmonary autograft.

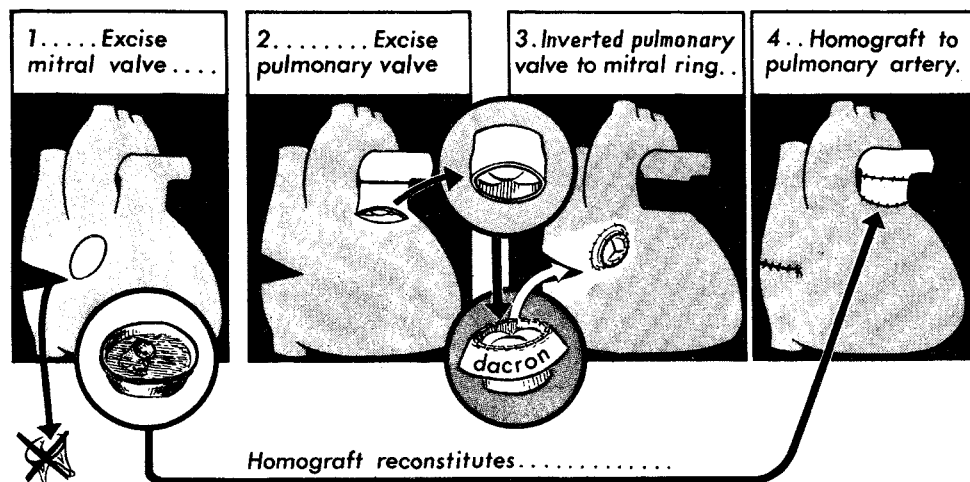


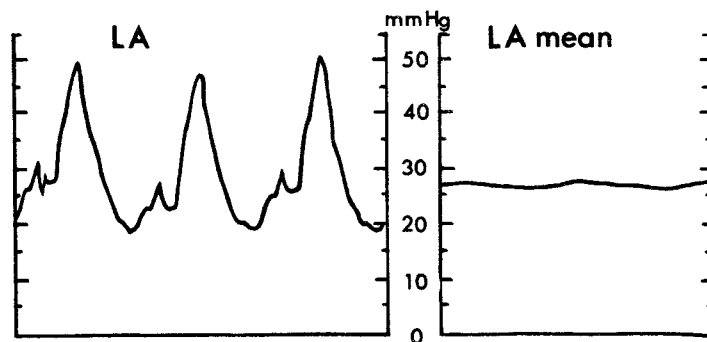
Fig. 4—Replacement of mitral valve with pulmonary autograft.

The pulmonary valve is then inserted into the aortic root in routine manner as for an aortic homograft (Ross 1967). This is followed by the insertion of the reconstituted aortic homograft in the right-ventricular outflow and its attachment proximally to the rim of excised right ventricular muscle and distally to the transected trunk of the pulmonary artery (figs. 2 and 3). Average perfusion-times have been two hours.

Mitral Replacement

Hubka et al. (1967) have described a method of substituting the inverted aortic homograft for the mitral valve and this has been applied clinically in 9 patients. Ionescu et al. (1967), too, have used aortic heterografts in this region. Following upon

PRE-OPERATIVE



POST-OPERATIVE

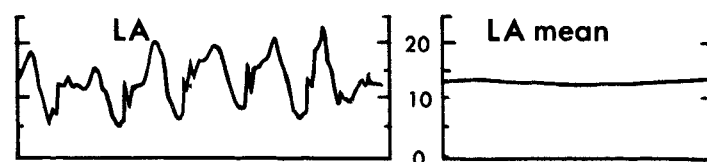


Fig. 6—Preoperative and postoperative left-atrial tracings after replacement of mitral valve with pulmonary autograft.

this, I have used pulmonary-valve autografts to replace the mitral valve in 2 patients, and there is further support for this technique in the experimental work reported by Lower et al. (1961). They inserted inverted pulmonary-valve autografts in the mitral area in dogs, finding viable pulmonary-valve tissue as long as nine months postoperatively.

An exactly similar technique is used for excision of the pulmonary valve. The rim of the pulmonary artery is then sewn to the residual annulus of the mitral valve in an inverted position and attached proximally to the wall of the left atrium by means of a small 'Dacron' ring. The excised pulmonary valve is again replaced with an aortic homograft (fig. 4).

Results

The technique of valve replacement with a pulmonary autograft has been used in 14 patients. 12 of these had the aortic valve replaced, with two deaths unrelated to the valve; one death was due to subarachnoid hæmorrhage, and one to a coronary embolism. The remaining 10 patients are well, with no evidence of regurgitation in either the aortic or pulmonary valve (figs. 5 and 6). The longest follow-up is now five months.

Both patients with pulmonary autografts in the mitral areas are well and with fully competent valves at three and two months.

Discussion

The initial results are encouraging, although the problem of the homograft has now been transferred to the pulmonary area. Further efforts to prevent the factors promoting calcification in homografts are needed. Nevertheless, the hope is that the reduced closing pressures in the pulmonary artery will also reduce the wear and tear on the cusps, and in any event dysfunction of the pulmonary valve is less likely to have fatal consequences for the patient. As an alternative, simple bicuspid pulmonary valves could be fashioned from autogenous pericardium or fascia lata.

Complete transection of the pulmonary artery does not entail great technical difficulties during bypass, and the approach provides excellent access to the main trunk of the left coronary artery for direct surgery.

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"... The fact that a substantial change in the expected pattern of births could start in 1956 without the reasons being in any way understood indicates one very great gap in our knowledge which might be filled by concentrated efforts by competent people... It is quite possible that the necessary basic knowledge is already to hand to enable us to regulate the population to fit its environment, and it may merely be a question of applying the scientific and other information which we have. But the problem will have to be looked at as a whole, and those primarily working on birth control will have to be closely linked with psychologists, agriculturists and the rest before any coherent policy can be expected to emerge."—Lord FLOREY, *The Responsibilities of Medicine in the Modern World*; p. 21. Royal College of Physicians of Edinburgh publication no. 32, 1967.

URINARY EXCRETION OF CALCIUM AND MAGNESIUM IN PATIENTS WITH CALCIUM-CONTAINING RENAL STONES

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Summary The 24-hour excretion of calcium and magnesium were determined in the urine of fifty-four male patients with calcium-containing renal stones. The calcium output was greater than 300 mg. per day in thirty-seven patients and the mean calcium excretion was significantly higher in the stone formers than in local control groups. The mean daily urinary magnesium excretion of the "normocalciuric" renal-stone formers (seventeen patients with daily urinary calcium excretion below 300 mg.) was significantly lower than that of normocalciuric controls, similarly defined. Four of these normocalciuric patients had very low 24-hour urinary magnesium outputs. The results support the view that low urinary magnesium excretion may be a factor in the production of renal calculi in some cases; the practical value of oral magnesium salts in preventing further stones in such patients is being investigated.

Introduction

DESPITE increasing interest in calcium-containing renal stones, the exact mechanisms of their pathogenesis remain largely obscure and their treatment is difficult and controversial. In some patients with renal stones an aetiological factor such as hyperparathyroidism, sarcoidosis, renal tubular acidosis, vitamin-D intoxication, primary hyperoxaluria, infection, or an anatomical abnormality of the renal tract can be demonstrated. Many other stone formers have hypercalciuria associated with a normal plasma-calcium and no other apparent cause; Albright et al. (1953) called this condition "idiopathic hypercalciuria".

Since the pioneer observations of Flocks (1939) it has usually been accepted that a high urinary calcium excretion predisposes to urinary calculi, although the exact mechanisms of renal-stone formation are poorly understood even in hypercalciuric individuals. Several workers have found in healthy populations numbers of symptomless individuals with high urinary calcium excretions (Hodgkinson and Pyrah 1958, Jackson and Dancaaster 1959, Watson and Dale 1966). We think that hypercalciuria is one of several factors which may contribute to renal-stone formation and we have found the reduction of urinary calcium output to be a good treatment in hypercalciuric renal-stone formers (Dent and Watson 1965), no matter what other factors may be concerned.

Many renal-stone formers have normal urinary calcium excretion, and the search continues for other aetiological factors. Some clinical and experimental evidence suggests that a low urinary excretion of magnesium may lead to renal-stone formation in some patients. In support of this view we describe here the results obtained from a study of 24-hour urinary excretion of calcium and magnesium in fifty-four male patients with calcium-containing renal

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